

BG770A-GL&BG95xA-GL

QuecLocator Application Note

LPWA Module Series

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About the Document

Revision History

Version	Date	Author	Description
-	2021-11-17	Jelena JOVANOVIC	Creation of the document
1.0	2022-01-14	Jelena JOVANOVIC/ Nikola BLAZIC	First official release

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1 Introduction

This application note introduces the AT commands defined for the QuecLocator® service supported by Quectel BG770A-GL, BG950A-GL and BG951A-GL modules.

QuecLocator® is an efficient positioning technology developed by Quectel. By integrating the cellular network information, it boosts the stand-alone GNSS performance, especially in challenging environments, such as urban canyons, indoors, parking garages, areas under overpasses, or in case the GNSS signal is blocked or intermittent. For more details about QuecLocator®, visit [Quectel IoT Portal](#).

NOTE

QuecLocator® is a value-added function provided by Quectel. For more details about the service, contact Quectel sales representatives or Technical Support (support@quectel.com).

2 QuecLocator[®] Authentication

To use the QuecLocator service, an authorization token is necessary for identity authentication. The auth token remains valid after the module reboots or powers off. Execute **AT+QLBSCFG="token"** to confirm whether a token has already been configured when you use the service for the first time; if not, please input one. The general steps are illustrated in the flowchart below.

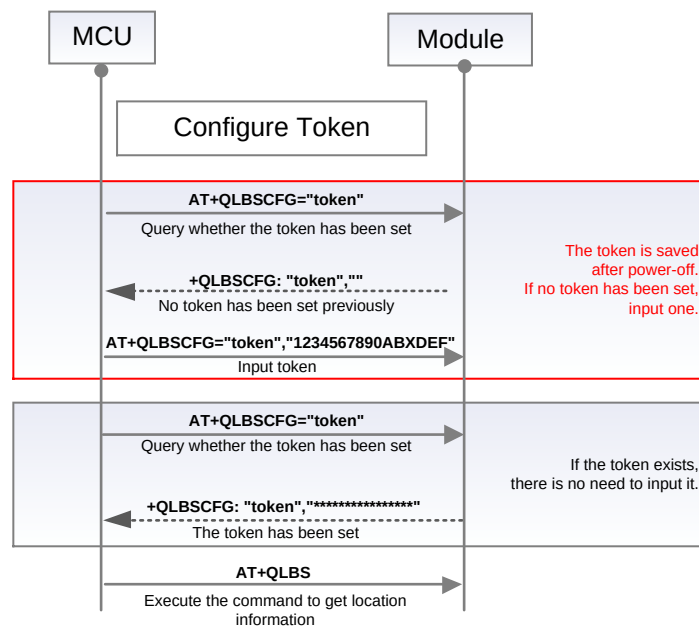


Figure 1: QuecLocator[®] Auth Token Configuration

NOTE

Consult Quectel Technical Support to apply for the token.

3 QuecLocator® AT Commands

3.1. AT Command Introduction

3.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

3.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 1: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of corresponding command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of a corresponding Write Command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

3.2. Declaration of AT Command Examples

The AT command examples in this document are provided to help you learn about the use of the AT commands introduced herein. The examples, however, should not be taken as Quectel's recommendations or suggestions about how to design a program flow or what status to set the module into. Sometimes multiple examples may be provided for one AT command. However, this does not mean that there is a correlation among these examples, or that they should be executed in a given sequence.

3.3. Description of QuecLocator AT Commands

3.3.1. AT+QLBSCFG Configure Parameters for QuecLocator

This command configures QuecLocator related parameters.

AT+QLBSCFG Configure Parameters for QuecLocator	
Test Command AT+QLBSCFG=?	Response +QLBSCFG: "asynch",(list of supported <asynch_mode>s) +QLBSCFG: "timeout",(range of supported <response_time>s) +QLBSCFG: "server",<server_name> +QLBSCFG: "token",<token_value> +QLBSCFG: "timeupdate",(list of supported <update_mode>s) +QLBSCFG: "withtime",(list of supported <time_mode>s) +QLBSCFG: "latorder",(list of supported <order_mode>s) +QLBSCFG: "scanband",(list of supported <scan_mode>s),(list of supported <scan_band>s) +QLBSCFG: "singlecell",(list of supported <singlecell_mode>s) OK
Read Command AT+QLBSCFG?	Response +QLBSCFG: "asynch",<asynch_mode> +QLBSCFG: "timeout",<response_time> +QLBSCFG: "server",<server_name> +QLBSCFG: "token",<token_value> +QLBSCFG: "timeupdate",<update_mode> +QLBSCFG: "withtime",<time_mode> +QLBSCFG: "latorder",<order_mode> +QLBSCFG: "scanband",<scan_mode>[,<scan_band>] +QLBSCFG: "singlecell",<singlecell_mode> OK

Write Command AT+QLBSCFG="asynch",<asynch_mode>]	Response If the optional parameter is omitted, query the current setting: +QLBSCFG: "asynch",<asynch_mode> OK If the optional parameter is specified, configure the execution mode of AT+QLBS: OK If there is an error related to ME functionality: +CME ERROR: <result>
Write Command AT+QLBSCFG="timeout",<response_time>]	Response If the optional parameter is omitted, query the current setting: +QLBSCFG: "timeout",<response_time> OK If the optional parameter is specified, configure the maximum response time of AT+QLBS: OK If there is an error related to ME functionality: +CME ERROR: <result>
Write Command ¹ AT+QLBSCFG="server",<server_name>]	Response If the optional parameter is omitted, query the current setting: +QLBSCFG: "server",<server_name> OK If the optional parameter is specified, configure the domain name/IP address and port number of the server that provides the positioning service: OK If there is an error related to ME functionality: +CME ERROR: <result>
Write Command AT+QLBSCFG="token",<token_value>]	Response If the optional parameter is omitted, query the current setting: (1) If <token_value> has been set previously: +QLBSCFG: "token", "*****"

¹ Usually, there is no need to configure **<server_name>**. When a non-default server is used for positioning service, configure the domain name/IP address and port number of the server with this command.

	OK (2) If <token_value> has not been set previously: +QLBSCFG: "token", "" OK If the optional parameter is specified, configure the token value for server authentication: OK If there is an error related to ME functionality: +CME ERROR: <result>
Write Command AT+QLBSCFG="timeupdate",<update_mode>]	Response If the optional parameter is omitted, query the current setting: +QLBSCFG: "timeupdate",<update_mode> OK If the optional parameter is specified, configure whether to update to the RTC the time acquired at accessing the server that provides the positioning service: OK If there is an error related to ME functionality: +CME ERROR: <result>
Write Command AT+QLBSCFG="withtime",<time_mode>]	Response If the optional parameter is omitted, query the current setting: +QLBSCFG: "withtime",<time_mode> OK If the optional parameter is specified, configure whether to output <time> in the response of AT+QLBS: OK If there is an error related to ME functionality: +CME ERROR: <result>
Write Command AT+QLBSCFG="latorder",<order_mode>]	Response If the optional parameter is omitted, query the current setting: +QLBSCFG: "latorder",<order_mode> OK

	If the optional parameter is specified, configure the latitude and longitude output order in the location information acquired with AT+QLBS: OK If there is an error related to ME functionality: +CME ERROR: <result>
Write Command AT+QLBSCFG="scanband"[,<scan_mode>[,<scan_band>]]	Response If the optional parameters are omitted, query the current settings: +QLBSCFG: "scanband",<scan_mode>[,<scan_band>] OK If any of the optional parameters is specified, configure the scan mode and the eMTC frequency band to be scanned. <scan_band> can be specified only when <scan_mode>=1. OK If there is an error related to ME functionality: +CME ERROR: <result>
Write Command AT+QLBSCFG="singlecell"[,<singlecell_mode>]	Response If the optional parameter is omitted, query the current setting: +QLBSCFG: "singlecell",<singlecell_mode> OK If the optional parameter is specified, enable or disable cellular positioning (positioning based on cell ID with algorithm): OK If there is an error related to ME functionality: +CME ERROR: <result>
Maximum Response Time	300 ms
Characteristics	The commands take effect immediately. The configurations will not be saved except those of <asynch_mode> and <token_value>.

Parameter

<asynch_mode>	Integer type. Command execution mode of AT+QLBS. In different modes, the response of the command varies. See Chapter 3.3.2 for details. <u>0</u> Synchronous mode (Not supported in eMTC network) 1 Asynchronous mode
----------------------------	--

<response_time>	Integer type. Maximum response time of AT+QLBS . If there is no response from the server within <response_time> , the command times out. Range: 10–120. Default: 60. Unit: second.																																
<server_name>	String type. Domain name/IP address and port number of the server. Format: "domain name/IP address:port" (e.g. "www.queclocator.com:80"). Port number range: 1–65535. Default value: "www.queclocator.com:80".																																
<token_value>	String type. Value of the authentication token. Length: 16 bytes.																																
<update_mode>	Integer type. Whether to update to the RTC the time acquired at accessing the server that provides the positioning service. <u>0</u> Do not update the time to RTC 1 Update the time to RTC																																
<time_mode>	Integer type. Whether to output <time> in the response of AT+QLBS . <u>0</u> Do not output 1 Output																																
<order_mode>	Integer type. Latitude and longitude output order in the location information acquired with AT+QLBS . 0 Output longitude before latitude: +QLBS: <loc_result>,<longitude>,<latitude>[,<time>] <u>1</u> Output latitude before longitude: +QLBS: <loc_result>,<latitude>,<longitude>[,<time>] See Chapter 3.3.2 for details of the parameters.																																
<scan_mode>	Integer type. Scan mode. <u>0</u> Scan the currently attached band 1 Scan full bands or a specified band																																
<scan_band>	A hexadecimal value that specifies the eMTC frequency band to be scanned. For example, if it is set to 0x40000000, LTE B31 will be scanned. This parameter can be configured only when <scan_mode> =1. <table data-bbox="422 1317 1321 1973"> <tbody> <tr><td>0x1 (BAND_PREF_LTE_BAND1)</td><td>LTE B1</td></tr> <tr><td>0x2 (BAND_PREF_LTE_BAND2)</td><td>LTE B2</td></tr> <tr><td>0x4 (BAND_PREF_LTE_BAND3)</td><td>LTE B3</td></tr> <tr><td>0x8 (BAND_PREF_LTE_BAND4)</td><td>LTE B4</td></tr> <tr><td>0x10 (BAND_PREF_LTE_BAND5)</td><td>LTE B5</td></tr> <tr><td>0x80 (BAND_PREF_LTE_BAND8)</td><td>LTE B8</td></tr> <tr><td>0x800 (BAND_PREF_LTE_BAND12)</td><td>LTE B12</td></tr> <tr><td>0x1000 (BAND_PREF_LTE_BAND13)</td><td>LTE B13</td></tr> <tr><td>0x20000 (BAND_PREF_LTE_BAND18)</td><td>LTE B18</td></tr> <tr><td>0x40000 (BAND_PREF_LTE_BAND19)</td><td>LTE B19</td></tr> <tr><td>0x80000 (BAND_PREF_LTE_BAND20)</td><td>LTE B20</td></tr> <tr><td>0x1000000 (BAND_PREF_LTE_BAND25)</td><td>LTE B25</td></tr> <tr><td>0x2000000 (BAND_PREF_LTE_BAND26)</td><td>LTE B26</td></tr> <tr><td>0x4000000 (BAND_PREF_LTE_BAND27)</td><td>LTE B27</td></tr> <tr><td>0x8000000 (BAND_PREF_LTE_BAND28)</td><td>LTE B28</td></tr> <tr><td>0x2000000000000000 (BAND_PREF_LTE_BAND66)</td><td>LTE B66</td></tr> </tbody> </table>	0x1 (BAND_PREF_LTE_BAND1)	LTE B1	0x2 (BAND_PREF_LTE_BAND2)	LTE B2	0x4 (BAND_PREF_LTE_BAND3)	LTE B3	0x8 (BAND_PREF_LTE_BAND4)	LTE B4	0x10 (BAND_PREF_LTE_BAND5)	LTE B5	0x80 (BAND_PREF_LTE_BAND8)	LTE B8	0x800 (BAND_PREF_LTE_BAND12)	LTE B12	0x1000 (BAND_PREF_LTE_BAND13)	LTE B13	0x20000 (BAND_PREF_LTE_BAND18)	LTE B18	0x40000 (BAND_PREF_LTE_BAND19)	LTE B19	0x80000 (BAND_PREF_LTE_BAND20)	LTE B20	0x1000000 (BAND_PREF_LTE_BAND25)	LTE B25	0x2000000 (BAND_PREF_LTE_BAND26)	LTE B26	0x4000000 (BAND_PREF_LTE_BAND27)	LTE B27	0x8000000 (BAND_PREF_LTE_BAND28)	LTE B28	0x2000000000000000 (BAND_PREF_LTE_BAND66)	LTE B66
0x1 (BAND_PREF_LTE_BAND1)	LTE B1																																
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0x2000000000000000 (BAND_PREF_LTE_BAND66)	LTE B66																																
<singlecell_mode>	Integer type. Enable or disable cellular positioning.																																

	0	Disable
	1	Enable
<result>	Integer type. Result code. See Chapter 5 for more information.	

NOTE

1. The **<token_value>** is used for server authentication. Apply to Quectel for it in advance.
2. The **<scan_mode>** is only supported in LTE-M (eMTC) network. The default band to be scanned is the one to which the module has attached. If **<scan_mode>=1** and **<scan_band>** is not specified, **AT+QLBS** is executed to scan full bands and you need to wait for more than 1 minute for the response.
3. When you configure **<scan_band>**, make sure the specified band is supported by the current operator. Otherwise, the module cannot get cell information of the band.

3.3.2. AT+QLBS Get Location Information by QuecLocator

This command outputs the location information by QuecLocator.

AT+QLBS Get Location Information by QuecLocator

Test Command	Response
AT+QLBS=?	OK
Execution Command	Response
Output the location information with cellular positioning enabled.	If the module is in synchronous mode (<asynch_mode>=0) and the positioning succeeds:
AT+QLBS	+QLBS: <loc_result>,<latitude>,<longitude>[,<time>]
	OK
	If the module is in synchronous mode and the positioning fails:
	+QLBS: <loc_result>
	OK
	If the module is in asynchronous mode (<asynch_mode>=1) and the positioning succeeds:
	OK
	+QLBS: <loc_result>,<latitude>,<longitude>[,<time>]
	If the module is in asynchronous mode and the positioning fails:
	OK
	+QLBS: <loc_result>

	If there is an error related to ME functionality: +CME ERROR: <result>
Maximum Response Time	Depends on the <response_time> in AT+QLBSCFG="timeout", <response_time> .
Characteristics	The command takes effect immediately. The configurations will not be saved.

Parameter

<latitude>	Float type. Latitude of the location information. This value is accurate to six decimal places. Range: -90.000000 to 90.000000.
<longitude>	Float type. Longitude of the location information. This value is accurate to six decimal places. Range: -180.000000 to 180.000000.
<time>	String type. Date and time obtained from HTTP header.
<loc_result>	Integer type. Positioning result. 0 Positioning succeeded Other values Positioning failed (See Chapter 5 for more information.)
<result>	Integer type. Result code. See Chapter 5 for more information.

4 Example

```
//Activate network first, then make configurations and get locations referring to the steps below.
AT+QLBSCFG="token" //Query the token value for authentication required
                    //by QuecLocator server.
+QLBSCFG: "token", "" //No token has been set previously.

OK
AT+QLBSCFG="token", "1234567890ABCDEF" //Input the token value for authentication required
                                        //by QuecLocator server.

OK
AT+QLBSCFG="token" //Query the token value for authentication required
                  //by QuecLocator server.
+QLBSCFG: "token", "*****"

OK
AT+QLBSCFG="latorder",1 //Configure the location output format into:
                        +QLBS: <loc_result>,<latitude>,<longitude>

OK
AT+QLBSCFG="asynch",1 //Configure the execute mode of AT+QLBS to
                      //asynchronous mode.

OK
AT+QLBS //Start cellular positioning.

OK

+QLBS: 0,31.847649,117.200134
```

5 Summary of Error Codes

Table 2: Summary of Error Codes

Code of <result>	Description of Error Codes
10000	QuecLocator fix failed
10001	Illegal IMEI number
10002	Token does not exist
10003	The number of devices using the same token exceeds the limit
10004	Times of positioning initiated by the same device in one day exceeds the limit
10005	Total times of positioning using the same token exceeds the limit
10006	Token expired
10007	IMEI number not accepted by the server
10008	Times of positioning using the same token within one day exceeds the limit
10009	The frequency of positioning using the same token exceeds the limit

NOTE

QuecLocator uses HTTP protocol. If any HTTP error code is returned, see [Quectel_BG770A-GL&BG95xA-GL_HTTP\(S\)_Application_Note](#) for the description of the error code.

6 Appendix Reference

Table 3: Terms and Abbreviations

Abbreviation	Description
eMTC	Enhanced Machine Type of Communication
GNSS	Global Navigation Satellite System
HTTP	Hypertext Transfer Protocol
IMEI	International Mobile Equipment Identity
IoT	Internet of Things
IP	Internet Protocol
LPWA	Low-Power Wide-Area
LTE-M	Long Term Evolution for Machines
MCU	Microcontroller Unit
ME	Mobile Equipment
RTC	Real-Time Clock
TA	Terminal Adapter
URC	Unsolicited Result Code